

Introduction to Computer Science

Spring 2026

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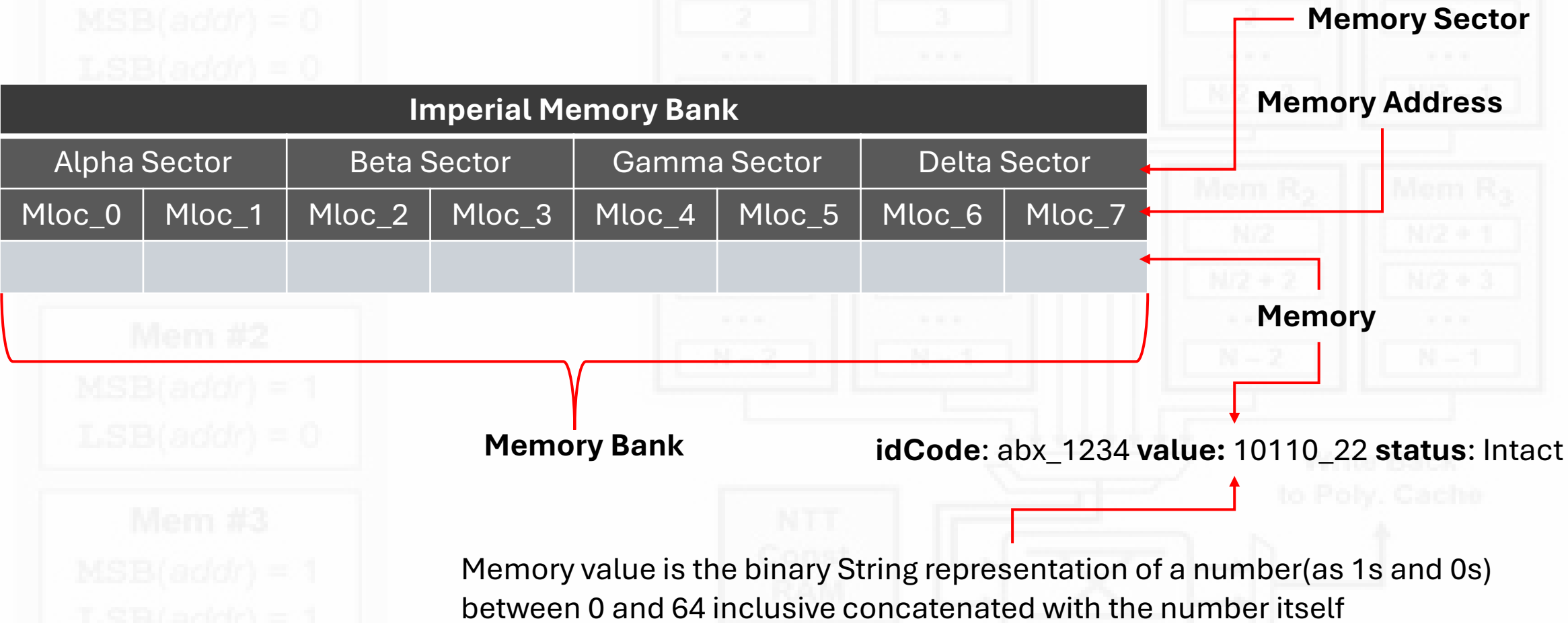


Homework 6

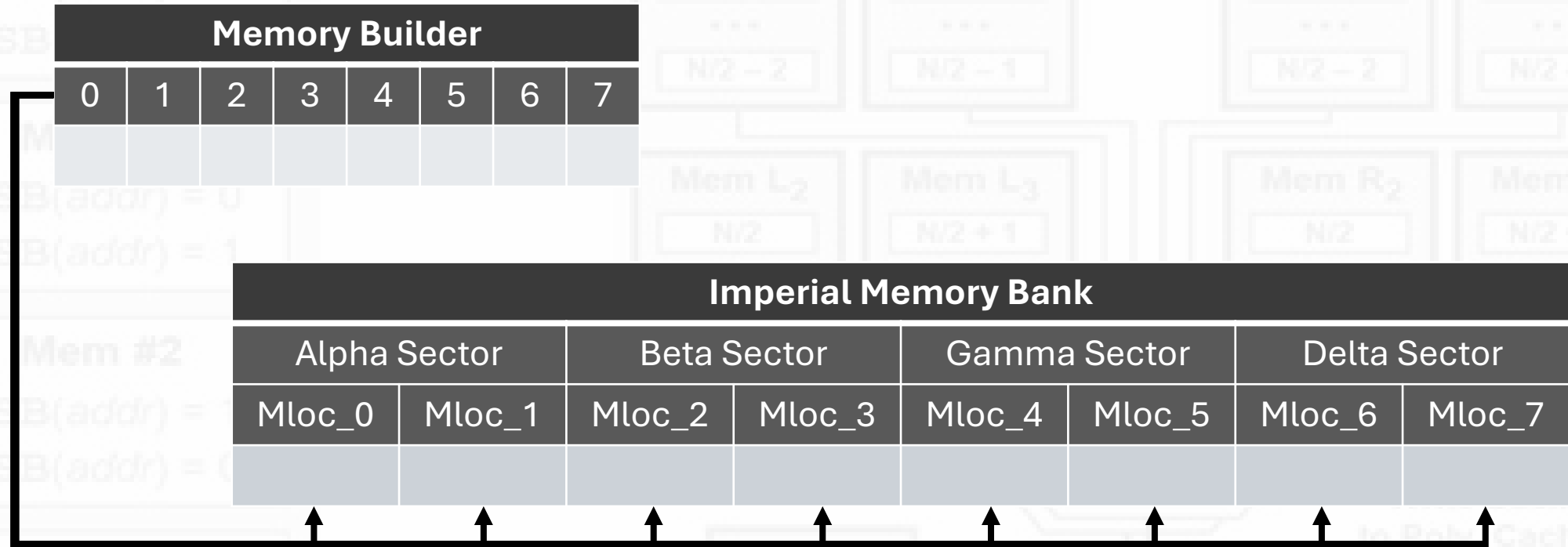
Professor HG Locklear

Imperial Memory Bank Structure

Imperial Memory Bank							
Alpha Sector		Beta Sector		Gamma Sector		Delta Sector	
Mloc_0	Mloc_1	Mloc_2	Mloc_3	Mloc_4	Mloc_5	Mloc_6	Mloc_7

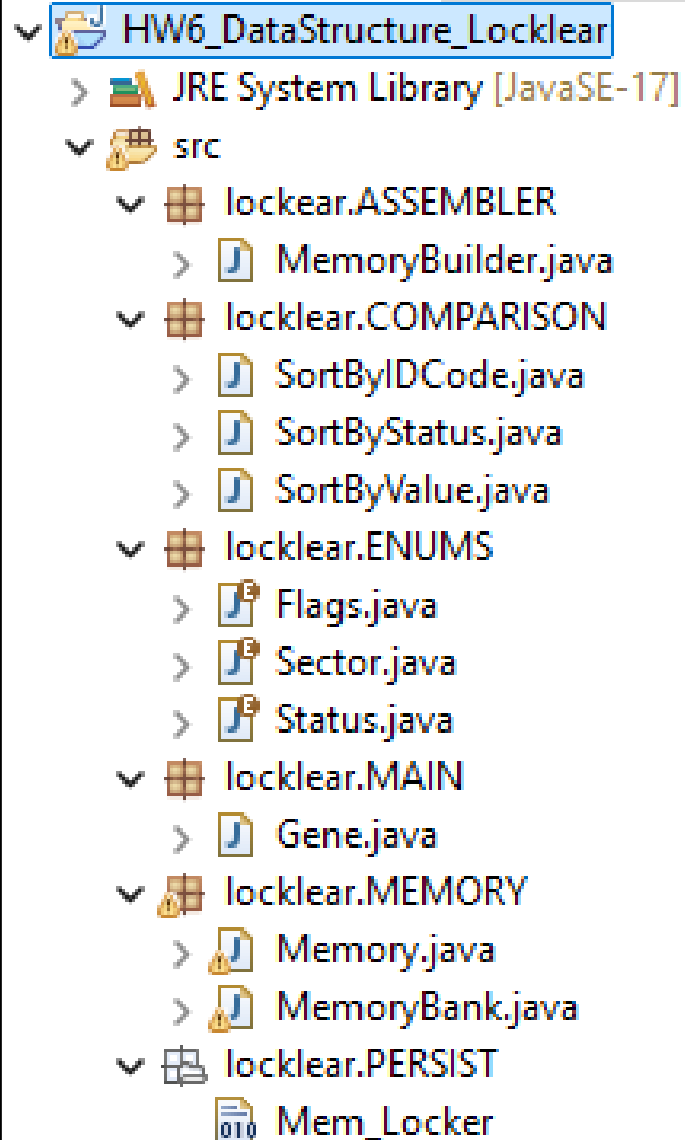


Imperial Memory Bank Construction



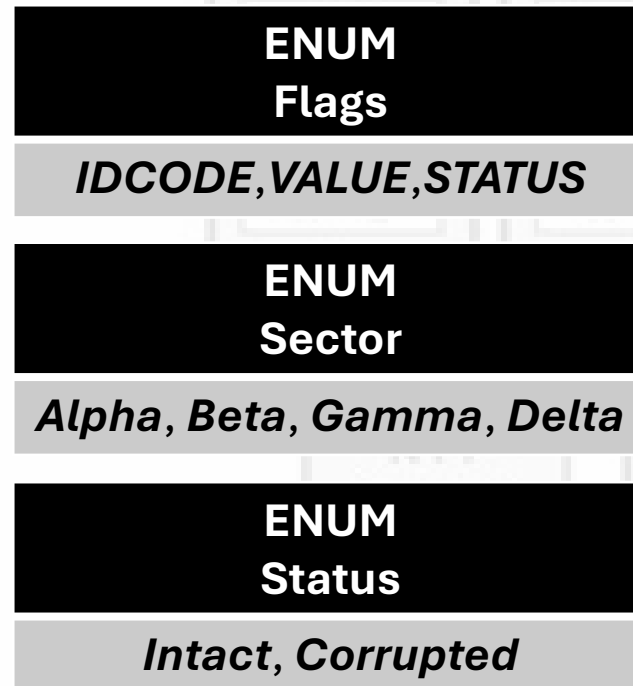
Memory is first constructed by the Memory Builder and then loaded into the Memory Bank

Program Structure



Create your project and program structure EXACTLY as specified.

Use the provided file for your main method (**Gene Class**)



(b)

Memory

Memory
-String idCode -String value -Status status
Memory() Memory(String idCode, String value, Status status)
+toString(): String +display(): void +fix(): boolean +erase(): boolean

TreeMap<String, String>

<https://docs.oracle.com/javase/8/docs/api/java/util/TreeMap.html>

The **no argument constructor** initializes no state.

If the **idCode**, **value**, and **status** of a Memory object are all null the Memory object is null

(a)

(b)

Memory

Memory Method Specification Chart			
Method	Input	Processing	Output
toString	None	Returns a String representation of the Memory object in the format shown below	String
display	None	Prints to the console the String returned by the toString method	Console
fix	None	Sets the status of the Memory object to Status.Intact if it is currently Status.Corrrupted and returns true	None
erase	None	Sets the value of idCode , value , and status to null and returns true	None

Memory: ypk 3530 Value: 10110 22 Status: Intact

MemoryBank

ArrayList<Memory>

<https://docs.oracle.com/javase/8/docs/api/java/util/ArrayList.html>

MemoryBank

-int bankID

MemoryBank()

MemoryBank(String bankID, MemoryBuilder M)

+display: void

+flush(): boolean

+dump(): Memory[]

+replicate(): MemoryBank

+reorder(Flags f): boolean

+displaySector(Sector s): void

+findMemory(String idCode): boolean

+returnMemory(String addr): Memory

+swapMemory(String addr1,String addr2): boolean

+getSector(Sector s): Memory[]

+insertMemory(Memory m): boolean

+insertCache(Memory[] m): boolean

+removeMemory(Memory[] m): boolean

+updateMemory(String addr, String value):boolean

+validateMemory(String[] addr): boolean

+persistMemory(String filename): void

+constituteMemory(String filename):MemoryBank

+compareBanks(MemoryBank mb): boolean

+repairMemory(): boolean

(a)

(b)

MemoryBank

MemoryBank Method Specification Chart

Method	Input	Processing	Output
display	None	<u>Prints</u> to the Console the bankID of the MemoryBank object and calls the display method for each Memory object in the MemoryBank. This method must use an iterator	Console
flush()	None	<u>Removes</u> all Memory objects from the MemoryBuilder and Prints to the Console ‘Memory Bank is [MemoryBank size] [MemoryBank size] / 8 Full’ and <u>returns</u> true	Boolean Console
dump()	None	<u>Returns</u> a 1D array of all Memory objects in the MemoryBuilder. Prints to the Console a String representation of the returned array showing the Memory object idCodes.	Memory[] Console
replicate()	None	<u>Returns</u> a copy of the MemoryBuilder object and calls the display method on the MemoryBuilder copy	MemoryBank Console
reorder	f	<u>Sorts</u> the MemoryBuilder based on the sort criteria f (which is a Memory value). <u>Returns</u> true	Boolean
displaySector	s	<u>Prints</u> to the Console the Sector and the Memory object values associate with that Sector (See <i>Expected Output for format</i>)	Console
findMemory	idCode	<u>Returns</u> the Memory object in the MemoryBuilder which has the idCode specified and Prints to the Console ‘Memory [addr] Found At Mloc_ [index where found]’ and. <u>returns</u> true if the Memory object is found ...and <u>returns</u> false and Prints to the Console ‘Memory Not Found’ if the MemoryBuilder object does not contain the specified Memory object	Boolean Console
returnMemory	addr	<u>Returns</u> the Memory object specified by addr	

MemoryBank

MemoryBank Method Specification Chart

Method	Input	Processing	Output
insertMemory	m	If the MemoryBank object contains less than 8 Memory objects, add the specified Memory object m and <u>returns</u> true. Otherwise, Prints to the Console ‘Memory Bank is Full Insert Fails’ and <u>returns</u> false	Boolean
insertCache	m	<u>Checks</u> that m length is not 0 and if so, <u>returns</u> false and Prints to the Console ‘Cache is Empty...No Memory Inserted’ otherwise <u>adds</u> all Memory objects in m to the MemoryBank Prints to the Console ‘Cache Inserted’ and <u>returns</u> true	Boolean Console
removeMemory	m	<u>Removes</u> any Memory object from the MemoryBank that corresponds to a Memory object in m and <u>returns</u> true	Boolean
updateMemory	addr value	<u>Sets</u> the Memory objected located at the specified Memory address (addr) to the value specified (value) and <u>returns</u> true. Prints to the Console “Memory Value was [old value] Memory Updated To [new value]” and <u>returns</u> true ...old and <i>new value must be accessed from the object</i>	Boolean Console
validateMemory	addr	Checks the specified Memory addresses in (addr) and Print to the Console how many of corresponding Memory objects are Status.Intact and Status.Corrupted in the format shown in the Expected Output Slide and <u>returns</u> true.	Boolean Console

MemoryBank

MemoryBank Method Specification Chart

Method	Input	Processing	Output
persistMemory	filename	<u>Serializes</u> the MemoryBank object to the specified file (filename). Prints to the Console <i>‘Memory Bank has been stored at [relative filepath to file]’</i>	Console
constituteMemory	filename	<u>Returns</u> the MemoryBank object stored in the specified file (filename) and calls the display method of the returned MemoryBank	MemoryBank Console
compareBanks	mb	<u>Compares</u> the MemoryBank (mb) to the MemoryBank object and Prints to the Console the results in the format shown on the Expected Output Slide and <u>returns</u> true. This method must use one or more iterators	Boolean Console
repairMemory	None	<u>Sets</u> the Status of any Memory object in the MemoryBank to Status.Intact whose status is Status.Corrupted and Prints to the Console <i>‘[repaired count] Memory Objects have been repaired’</i> (See Expected Output Slide) and <u>returns</u> true. This method must use an iterator.	Boolean Console
swapMemory	addr1 addr2	<u>Swaps</u> the Memory object located at addr1 to addr2 , and the Memory object located at addr2 to addr1	Boolean
getSector	s	<u>Returns</u> the Memory objects located in the specified Sector s	Memory[]

MemoryBuilder

LinkedBlockingQueue<Memory>

<https://docs.oracle.com/javase/8/docs/api/java/util/concurrent/LinkedBlockingQueue.html>

MemoryBuilder

-String id

MemoryBuilder(String id)

+generateIDCode(): String

+generateValue(int v): String

+generateStatus(): Status

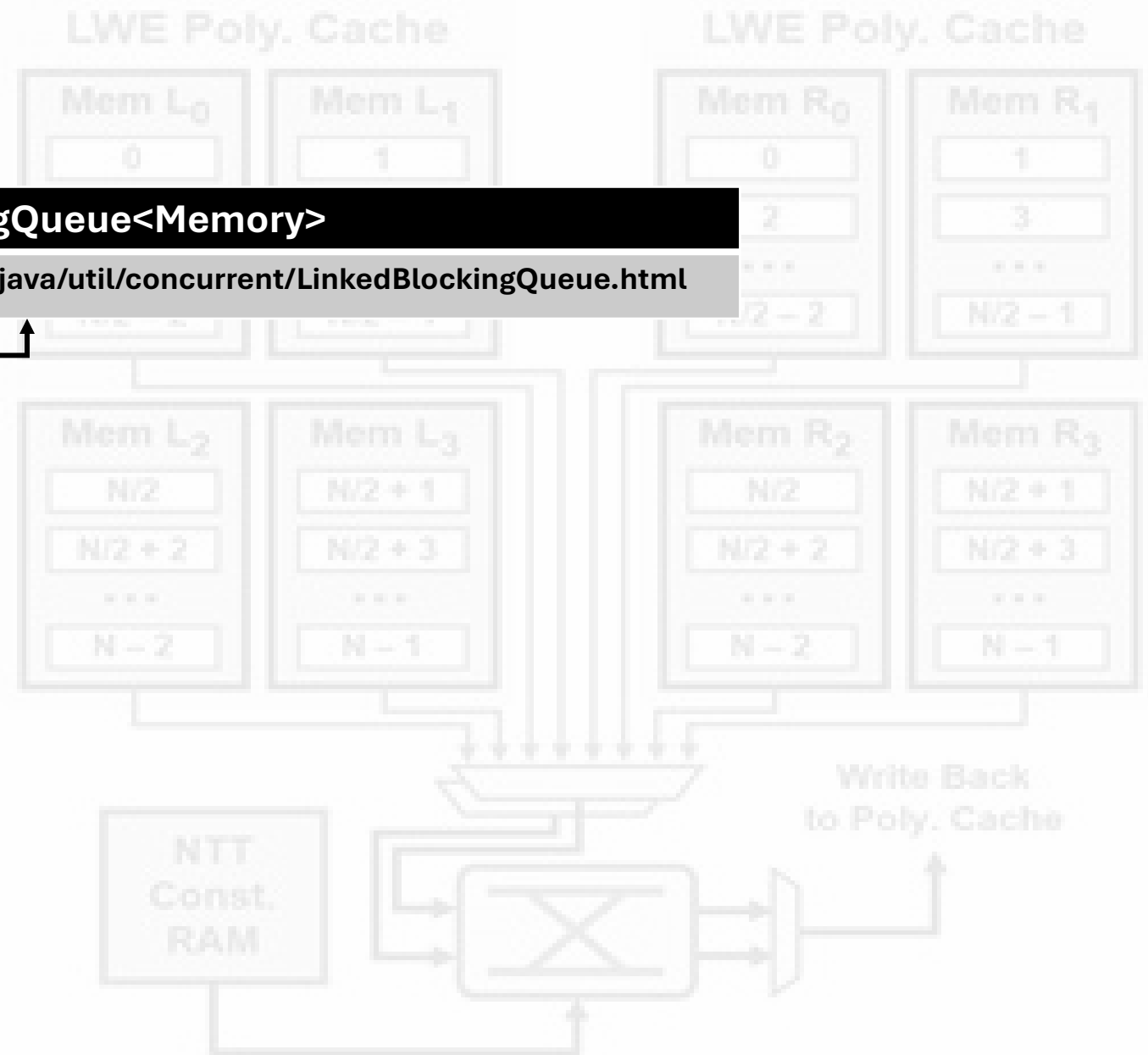
+buildMemory(): Memory

+offerMemory(Memory m): boolean

+pollMemory(): Memory

+clearMemory(): boolean

(a)



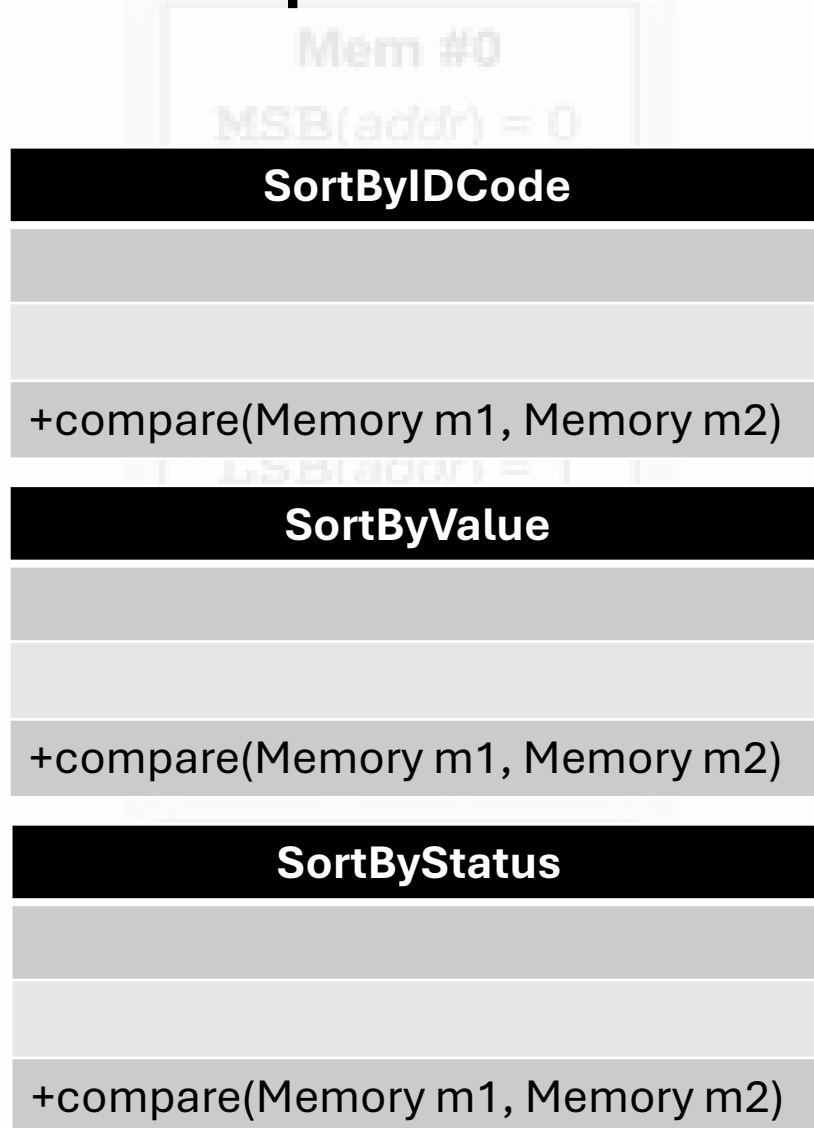
(b)

MemoryBuilder

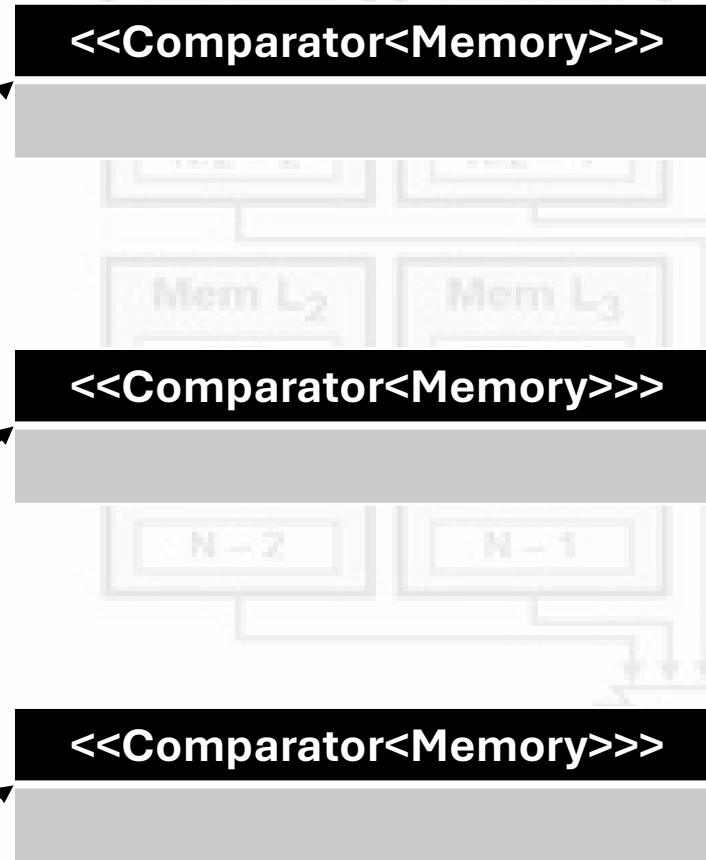
MemoryBuilder Method Specification Chart

Method	Input	Processing	Output
generateIDCode		Returns a String formatted as ***_#### where * is a randomly generated lowercase letter and # is a randomly generated digit (0-9)	String
generateValue	v	Returns a String formatted as a Memory value as specified on Slide 2 where v is a number between 0 and 64	String
generateStatus		Returns a randomly generated Status enumeration	Status
buildMemory		Randomly generates 8 Memory objects and stores them in this MemoryBuilder object. Prints 'Memory Build Complete' to the Console and returns true	Boolean Console
pollMemory		Returns the first Memory object in the MemoryBuilder object and Prints to the Console 'Memory Object [memory object] has been retrieved' .	Memory Console
offerMemory	m	If the MemoryBuilder object is not full (7 or less Memory objects) adds the Memory object m and Prints to the Console 'Memory Added' and returns true. Otherwise Prints to the Console 'Builder is Full' and returns false	Boolean
clearMemory		Removes every Memory object in the MemoryBuilder and Prints to the Console 'Memory Builder has [MemoryBuilder size] Memory Objects '	Boolean

Comparators



(a)



(b)

Each of the Comparator classes **implements their compare method** based on the attribute specified in the class name.

Expected Output Format Task 1-3

Memory Build Complete

TASK 1

Memory Bank ID: Bank_1

Memory: xqb_0427	Value: 10101_21	Status: Intact
Memory: mol_4032	Value: 110100_52	Status: Corrupted
Memory: awt_1157	Value: 101111_47	Status: Corrupted
Memory: lvk_0184	Value: 100_4	Status: Corrupted
Memory: fym_5456	Value: 101111_47	Status: Intact
Memory: lyo_2351	Value: 101101_45	Status: Corrupted
Memory: lmn_5974	Value: 110001_49	Status: Intact
Memory: cii_3739	Value: 11011_27	Status: Intact

TASK 2

Memory Bank is 0/8 Full

TASK 3

Memory Build Complete

Contains 8 Memory Objects

Expected Output Task 4-5

TASK 4

Memory Bank ID: Bank_2

Memory: bpy_1251	Value: 111_7	Status: Corrupted
Memory: mtr_5720	Value: 111100_60	Status: Intact
Memory: qwt_9078	Value: 11_3	Status: Corrupted
Memory: osd_1242	Value: 111001_57	Status: Corrupted
Memory: bvd_6056	Value: 10001_17	Status: Intact
Memory: rfk_2548	Value: 100011_35	Status: Intact
Memory: mfi_8547	Value: 100101_37	Status: Corrupted
Memory: rzz_1746	Value: 1101_13	Status: Corrupted

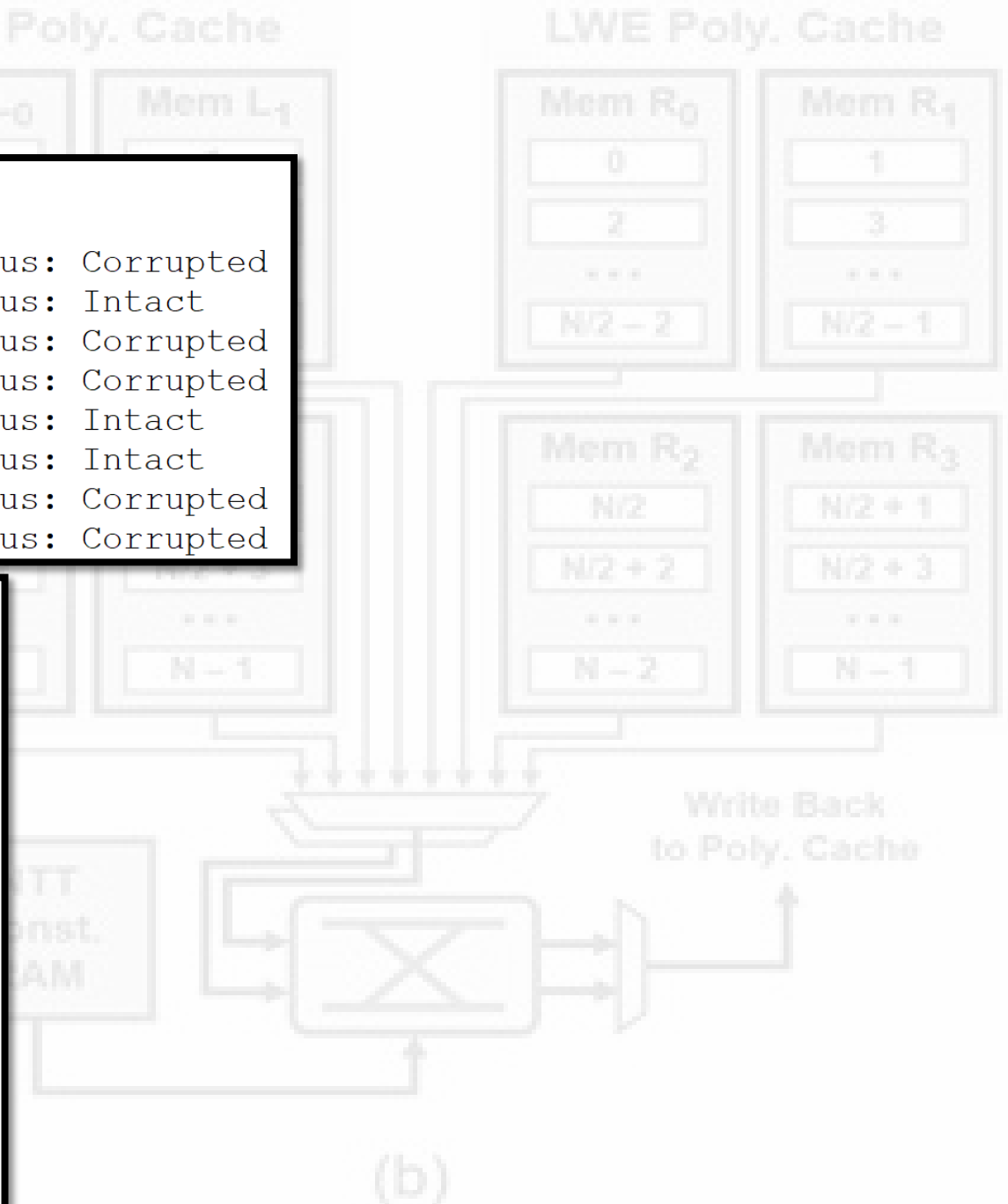
TASK 5

Memory Bank ID: Bank_2

Memory: bpy_1251	Value: 111_7	Status: Corrupted
Memory: mtr_5720	Value: 111100_60	Status: Intact
Memory: qwt_9078	Value: 11_3	Status: Corrupted
Memory: osd_1242	Value: 111001_57	Status: Corrupted
Memory: bvd_6056	Value: 10001_17	Status: Intact
Memory: rfk_2548	Value: 100011_35	Status: Intact
Memory: mfi_8547	Value: 100101_37	Status: Corrupted
Memory: rzz_1746	Value: 1101_13	Status: Corrupted

Memory Bank ID: Bank_2

Memory: bpy_1251	Value: 111_7	Status: Corrupted
Memory: qwt_9078	Value: 11_3	Status: Corrupted
Memory: osd_1242	Value: 111001_57	Status: Corrupted
Memory: mfi_8547	Value: 100101_37	Status: Corrupted
Memory: rzz_1746	Value: 1101_13	Status: Corrupted
Memory: mtr_5720	Value: 111100_60	Status: Intact
Memory: bvd_6056	Value: 10001_17	Status: Intact
Memory: rfk_2548	Value: 100011_35	Status: Intact



Expected Output Task 6-8

TASK 6

Sector :Alpha

Mloc_0: 111_7 Mloc_1: 11_3

TASK 7

Memory: bpy_1251 Value: 111_7 Status: Corrupted

TASK 8

Memory Bank ID: Bank_2

Memory: bpy_1251 Value: 111_7 Status: Corrupted

Memory: qwt_9078 Value: 11_3 Status: Corrupted

Memory: osd_1242 Value: 111001_57 Status: Corrupted

Memory: mfi_8547 Value: 100101_37 Status: Corrupted

Memory: rzz_1746 Value: 1101_13 Status: Corrupted

Memory: mtr_5720 Value: 111100_60 Status: Intact

Memory: bvd_6056 Value: 10001_17 Status: Intact

Memory: rfk_2548 Value: 100011_35 Status: Intact

Memory Bank ID: Bank_2

Memory: bpy_1251 Value: 111_7 Status: Corrupted

Memory: rfk_2548 Value: 100011_35 Status: Intact

Memory: osd_1242 Value: 111001_57 Status: Corrupted

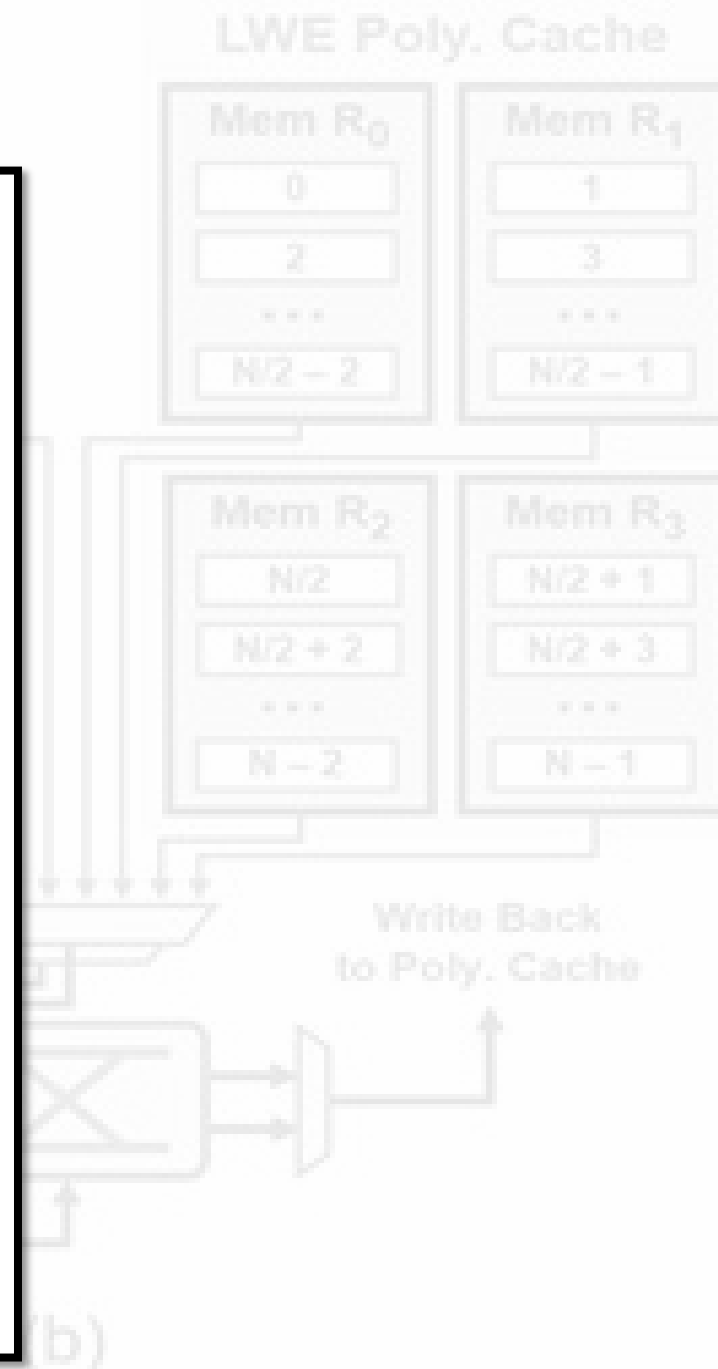
Memory: mfi_8547 Value: 100101_37 Status: Corrupted

Memory: rzz_1746 Value: 1101_13 Status: Corrupted

Memory: mtr_5720 Value: 111100_60 Status: Intact

Memory: bvd_6056 Value: 10001_17 Status: Intact

Memory: qwt_9078 Value: 11_3 Status: Corrupted



Expected Output Task 9-14

```
***TASK 9***  
[Memory: bpy_1251          Value: 111_7          Status: Corrupted,  
***TASK 10***  
Memory Bank ID: Bank_1  
Memory Bank ID: Bank_1  
Memory: abc_1234          Value: 10110_22          Status: Intact  
  
Memory Build Complete  
Memory Bank is Full Insert Fails  
***TASK 11***  
Memory Bank ID: null  
Cache Inserted  
Memory Bank ID: Bank_2  
Memory: abc_1234          Value: 10110_22          Status: Intact  
Memory: bbc_2234          Value: 10110_22          Status: Corrupted  
Memory: abc_3234          Value: 10110_22          Status: Intact  
***TASK 12***  
Memory Value was: 100101_37 Memory Updated To: 10110_22  
***TASK 13***  
Corrupted Memory: 0 Intact Memory: 2  
***TASK 14***  
Memory Bank has been stored at src\locklear\PERSIST\Mem_Locker
```

Expected Output Tasks 15-16

TASK 15

Memory Bank ID: Bank_2

Memory: bpy_1251	Value: 111_7	Status: Corrupted
Memory: rfk_2548	Value: 100011_35	Status: Intact
Memory: osd_1242	Value: 111001_57	Status: Corrupted
Memory: mfi_8547	Value: 10110_22	Status: Corrupted
Memory: rzz_1746	Value: 1101_13	Status: Corrupted
Memory: mtr_5720	Value: 111100_60	Status: Intact
Memory: bvd_6056	Value: 10001_17	Status: Intact
Memory: qwt_9078	Value: 11_3	Status: Corrupted

TASK 16

Memory Build Complete

joh_4700 <> lab_9079

ved_4858 <> ods_8822

oya_7725 <> enx_0151

hdh_7045 <> ygm_5073

wlf_7466 <> fir_7008

ymr_0908 <> sfj_7232

xja_5382 <> jgp_3476

ccq_7636 <> zwe_9836

Memory Banks Are Not Indentical

There are 8 Different Memory Object

joh_4700 = joh_4700

ved_4858 = ved_4858

oya_7725 = oya_7725

hdh_7045 = hdh_7045

wlf_7466 = wlf_7466

ymr_0908 = ymr_0908

xja_5382 = xja_5382

ccq_7636 = ccq_7636

Memory Banks Are Indentical

There are 0 Different Memory Object

(a)

(b)

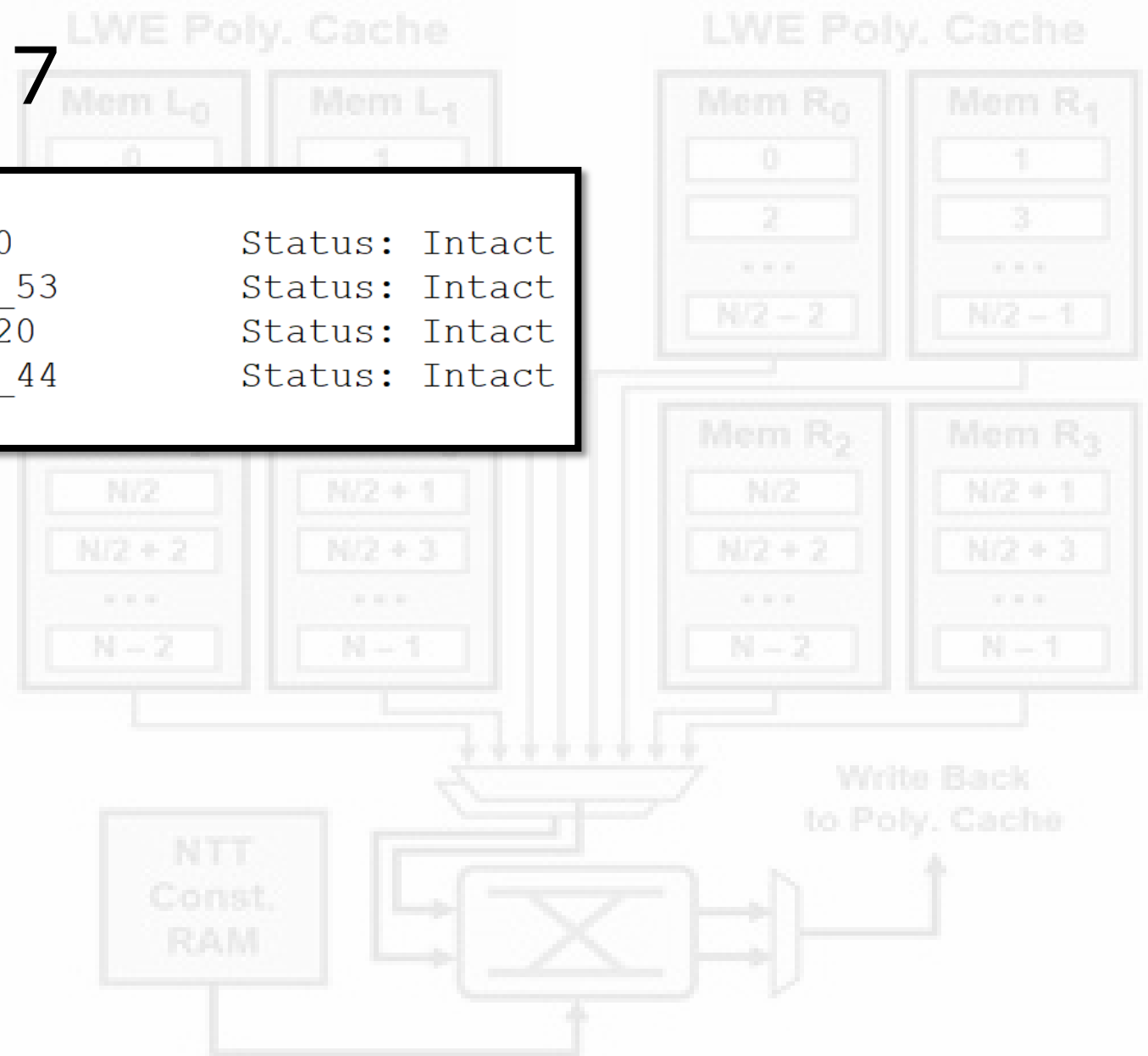
Expected Output Task 17

TASK 17

Memory: yxo_9902	Value: 1010_10	Status: Intact
Memory: mrj_3643	Value: 110101_53	Status: Intact
Memory: yps_0586	Value: 10100_20	Status: Intact
Memory: vrj_4696	Value: 101100_44	Status: Intact

4 Memory Objects have been repaired

(a)



(b)